

习题 3.4

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2025 年 7 月 29 日

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- (1)

令 $y = x - \frac{\pi}{2}$, $x \rightarrow \pi/2$ 时, $y \rightarrow 0$ 且 $t \neq 0$, 于是, 我们有

$$\begin{aligned}\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} &= \lim_{y \rightarrow 0} \frac{\cos(y + \frac{\pi}{2})}{y} \\ &= \lim_{y \rightarrow 0} \frac{-\sin(y)}{y} \\ &= -1\end{aligned}$$

- (8)

$$\begin{aligned}\lim_{x \rightarrow a} \frac{\sin^2(x) - \sin^2(a)}{x - a} &= \lim_{x \rightarrow a} \frac{[\sin(x) - \sin(a)][\sin(x) + \sin(a)]}{x - a} \\ &= \lim_{x \rightarrow a} \frac{2\cos(\frac{x+a}{2})\sin(\frac{x-a}{2})[\sin(x) + \sin(a)]}{x - a} \\ &= \lim_{x \rightarrow a} [\sin(x) + \sin(a)] \cos(\frac{x+a}{2}) \frac{\sin(\frac{x-a}{2})}{\frac{x-a}{2}} \\ &= 2\sin(a) \cdot \cos(a) \cdot 1 \\ &= \sin(2a)\end{aligned}$$

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- (4)

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{1+x}{1-x} \right)^{\frac{1}{x}} &= \lim_{x \rightarrow 0} \left(1 + \frac{2x}{1-x} \right)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \left(1 + \frac{2x}{1-x} \right)^{\frac{1-x}{2x} \cdot \frac{2x}{x(1-x)}}\end{aligned}$$

令 $t = \frac{2x}{1-x}$, 则 $x \rightarrow 0, t \rightarrow 0$, 所以

$$\begin{aligned}\lim_{x \rightarrow 0} \left(1 + \frac{2x}{1-x} \right)^{\frac{1-x}{2x}} &= e \\ \lim_{x \rightarrow 0} \frac{2x}{x(1-x)} &= 2\end{aligned}$$

利用 P80 例 1 可得:

$$\lim_{x \rightarrow 0} \left(1 + \frac{2x}{1-x} \right)^{\frac{1-x}{2x} \cdot \frac{2x}{x(1-x)}} = e^2$$

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说明 1. 这种极限套极限的格式, 按照运算的优先级进行计算。本题中, 先计算大括号里的极限, 把其中的 x 看做定值。

我们有

$$\begin{aligned}\sin x &= 2 \cos \frac{x}{2} \sin \frac{x}{2} \\ &= \cdots \\ &= 2^n \cos \frac{x}{2} \cos \frac{x}{2^2} \cdots \cos \frac{x}{2^n} \sin \frac{x}{2^n}\end{aligned}$$

于是,

$$\begin{aligned}\lim_{n \rightarrow \infty} \cos x \cos \frac{x}{2} + \cdots + \cos \frac{x}{2^n} &= \lim_{n \rightarrow \infty} \frac{\cos x \sin x}{2^n \sin \frac{x}{2^n}} \\ &= \lim_{n \rightarrow \infty} \frac{\cos x \sin x}{x} \cdot \frac{\frac{x}{2^n}}{\sin \frac{x}{2^n}} \\ &= \frac{\cos x \sin x}{x} \cdot 1 \\ &= \frac{\cos x \sin x}{x}\end{aligned}$$

所以

$$\lim_{x \rightarrow 0} \frac{\cos x \sin x}{x} = 1$$